X6EL, which is basically a SCSI scanner, took all of 41 seconds. The slowest scanner here was the Umax Astra 5400, a USB scanner; it took almost 46 seconds to scan the simple black and white document!

ScanMaker X12USL from Microtek took top honours in the high-end segment. It uses a SCSI interface and finished scanning the document in a neat 18 seconds. It was followed closely by the HP ScanJet 5470C, a USB scanner, which took a decent 19 seconds to scan the document. The rest took over 25 seconds, with the Umax Astra 4000U being the slowest—it took 37 seconds to finish the test.

So if you have tonnes of documents to scan, there is only one choice—the Microtek ScanMaker 3800. By the time you pick up the next page to scan, you will have the previous document scanned and ready to use!

Colour photo prescan speed: Most people preview a full photograph, crop it, select the areas they want and then scan it. This makes the time taken to preview very important, especially if you scan a truckload of photographs and graphically intensive documents. The scores in this test were quite varied and the preview times ranged from under 10 seconds to about 30-odd seconds. Ideally, a 10 second preview is what you should look for when scanning a photograph.

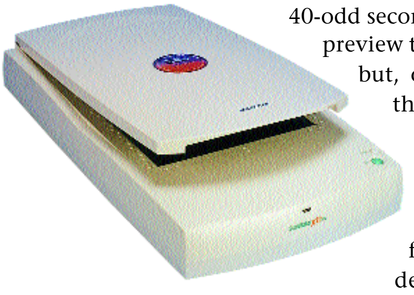
Most of the budget scanners took less than 20 seconds to preview. The Microtek ScanMaker 3800 performed well in this test too, taking 9 seconds to preview the photograph, but was surprisingly beaten by the Agfa SnapScan 310P, which came out with top honours here, taking 8 seconds to preview the photograph.

The scanners in the mid range segment were quite sluggish—not one scanner could break the 10 second barrier. The best times were logged by BenQ ScanPrisa 640BT (12 seconds) and the Umax Astra 4400 (13 seconds). The Microtek ScanMaker V6UPL was the slowest here, taking 32 seconds to finish the preview.

The high-end scanners got to show off their mettle in this test—the HP ScanJet 5470C and ScanJet 5400C finished the photo preview in 8 seconds flat. The slowest scanner here was the Umax Astra 4000U at a disappointing 25 seconds.

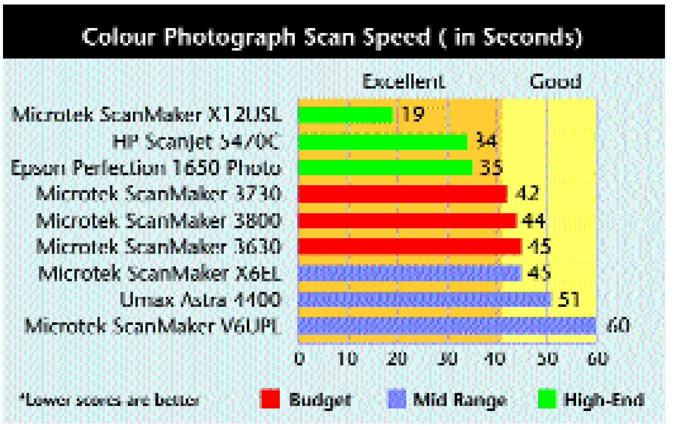
Colour photo scan speed: This is where speed really counts. Scanning full colour documents or photographs can really put the scanner, its interface and the TWAIN driver under severe stress. Besides, as most users buy a scanner to scan photographs, the time taken for this test becomes an important consideration.

We scanned a colour photograph at 300 dpi, and under ideal circumstances it should take not more than 40-odd seconds (multiplier of 4 on the preview time) to complete this test, but, on an average, we found the scanners taking about a minute.



Microtek ScanMaker X12USL offers blazing fast colour photo scanning speeds—our test photograph was scanned in an amazing 19 seconds!

A scanner with a faster interface such as SCSI or USB should be faster. This was clearly evident in the budget segment where the USB scanner from Canon (the CanoScan D646U) was significantly faster than its parallel port sibling, the Canon CanoScan N640P. The N640P took nearly 200 seconds (3 minutes 20 seconds) to scan the photograph, while the D646U took only 56 seconds, making it nearly three-and-a-half times as fast! For heavy scanning, a parallel interface truly becomes a bottleneck. The USB scanners on the other hand consistently logged a time of around 60 seconds. The Microtek ScanMaker 3730 was the fastest here, taking just 42 seconds to scan the colour photo. In fact, all the Microtek entries in the budget segment performed exceptionally well, taking around 40-45 seconds. The slowest USB scanner here was the Umax AstraSlim 600—it took over a minute-and-a-half to complete the test.



Once again, the scanners represented in the mid range segment did not live up to their price points. They were much slower than the budget 'speedos' and only the Microtek ScanMaker X6EL, a SCSI scanner, showed some promise by finishing the scan in 45 seconds. The rest turtled in at over a minute at an average.

The high-end segment proved to be quite exciting, with the photo being scanned at warp speeds! The Microtek ScanMaker X12USL was a true speed demon taking just 19 seconds to scan the photograph! The Microtek scanners performed well in the speed tests across segments, but this one truly takes the cake. With such scanning speed, you needn't even preview the photograph before scanning! Almost all the scanners finished the scanning in under our 40 second speed limit in this test, except for the Umax Astra 4000U which was surprisingly slow and took over a minute to scan the photograph.

This is the first time we've seen a test segment being completely dominated by a brand. Microtek seems to have gotten all the elements right when it comes to high speed scanning. Their scanners proved to be unbeatable across categories and are highly recommended for bulk scanning.

IT8 card test: Tonal quality is measured on a logarithmic (exponential) scale ranging from pure white to pure black. The tonal range is the difference between the darkest and the brightest optical densities (shades) a scanner can capture. The bigger the difference, the larger the dynamic range of the shades it can capture and hence better the scanner's image reproduction ability. The ability of a scanner to detect a wider tonal range is inherent in its hardware. A scanner that can detect sublime tonal variations needs to be built solidly from the bottom up. No amount of driver tweaking will make up for any build quality flaws.